

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250269745

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Yamamoto; Kozo

POWER UNIT ROOM INSTALLING STRUCTURE

Abstract

The power unit room installing structure includes: a power unit disposed opposite to the dash panel in the vehicle front-rear direction; a high voltage component disposed between the power unit and the dash panel; a charger disposed on the vehicle upper side of the power unit; a charging cable disposed from the charger toward the dash panel; and a regulating cable that is another cable different from the charging cable, wherein a part of the charging cable is covered from the vehicle upper side, thereby restricting movement of the charging cable toward the vehicle upper side and allowing movement toward at least the dash panel side.

Inventors: Yamamoto; Kozo (Nagoya-shi, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

Family ID: 1000008351133

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

Appl. No.: 18/975418

Filed: December 10, 2024

Foreign Application Priority Data

JP	2024-029117	Feb. 28, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: B60L53/18 (20190101); B60L1/02 (20060101); B60L3/00 (20190101)

U.S. Cl.:

CPC B60L53/18 (20190201); B60L1/02 (20130101); B60L3/0046 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-029117 filed on Feb. 28, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The disclosure relates to a power unit room installing structure.

2. Description of Related Art

[0003] A hybrid electric vehicle in which a charging cable, for supplying electric power from an external power source to a battery (charger) provided at a rear portion of a vehicle, is installed in a power unit room of the vehicle, has been conventionally known (e.g., see Japanese Unexamined Patent Application Publication No. 2010-241291 (JP 2010-241291 A)).

SUMMARY

[0004] However, in plug-in hybrid electric vehicles (PHEVs) and battery electric vehicles (EVs), which are chargeable, there are many components installed in the power unit room, and accordingly providing an empty space in the power unit room is difficult. Thus, it is desirable to provide means for suppressing occurrence of a secondary accident such as electric shock or the like, in preparation for when a high voltage component in the power unit room is damaged due to damage in the event of a collision.

[0005] Accordingly, an object of the disclosure is to provide a power unit room installing structure that is capable of suppressing occurrence of a secondary accident, in preparation for damage to a high voltage component in the power unit room, in the event of a collision.

[0006] A power unit room installing structure according to a first aspect of the disclosure includes [0007] a power unit that is installed in a power unit room of a vehicle, and that is disposed facing a dash panel separating the power unit room and a vehicle cabin in a vehicle front-rear direction, [0008] a high voltage component that is disposed between the power unit and the dash panel, [0009] a charger that is disposed on a vehicle upward side from the power unit, [0010] a charging cable that is routed from the charger toward the dash panel, and [0011] a regulating cable that is another cable different from the charging cable, and that is configured to, by covering a portion of the charging cable from the vehicle upward side, regulate movement of the charging cable toward the vehicle upward side, and allow movement of the charging cable toward at least the dash panel side.

[0012] In the power unit room installing structure according to the first aspect of the disclosure, the high voltage component is disposed between the power unit and the dash panel. The charging cable extending from the charger disposed on the vehicle upward side from the power unit is routed toward the dash panel. Also, the regulating cable covering the charging cable is disposed in the power unit room. The regulating cable is configured to regulate movement of the charging cable toward the vehicle upward side by covering part of the charging cable from the vehicle upward side, and allow movement of the charging cable toward at least the dash panel side.

[0013] Now, when the charging cable moves is when the vehicle is in a collision, for example. That is to say, when a collision load is input to the power unit room in the event of a collision of the vehicle, the charging cable moves together with the power unit and the charger in the power unit room. At this time, the charging cable is allowed to move toward the dash panel side by the restriction of the regulating cable, and accordingly moves between the power unit and the dash panel, whereby the high voltage component can be covered from the vehicle upward side. Thus, even when the high voltage component is damaged due to damage in the event of a collision, covering the high voltage component with the charging cable keeps a person who opens the power

unit room from easily coming into contact with the high voltage component, and accordingly occurrence of a secondary accident such as electric shock or the like is suppressed.

[0014] With the power unit room installing structure according to a second aspect of the disclosure, [0015] in the first aspect, the regulating cable is configured to allow movement of the charging cable toward the dash panel side such that the charging cable overlaps a live part of the high voltage component in plan view.

[0016] In the power unit room installing structure according to the second aspect of the disclosure, the charging cable is allowed to move so as to overlap the live part of the high voltage component by the regulating cable. Accordingly, a person who opens the power unit room can be effectively suppressed from coming into contact with the live part regarding which there is a high risk of electric shock.

[0017] With the power unit room installing structure of the third aspect according to the disclosure, in the first aspect or the second aspect, [0018] the regulating cable is a cable in which an extra length of the cable is set to be shorter than an extra length of the charging cable.

[0019] In the power unit room installing structure according to the third aspect of the disclosure, the regulating cable is a cable in which an extra length of the cable is set to be shorter than an extra length of the charging cable, and accordingly the movement of the charging cable to the vehicle upper side can be readily regulated.

[0020] With the power unit room installing structure according to a fourth aspect of the disclosure, in any one aspect of the first aspect to the third aspect, the regulating cable is a cable that connects a transaxle included in the power unit, and the charger.

[0021] In the power unit room installing structure according to the fourth aspect of the disclosure, [0022] the cable connecting the transaxle and the charger is used as the regulating cable.

[0023] The cable connecting the transaxle and the charger is usually made up of a high voltage harness, and accordingly the cable has a relatively large cross-sectional area, and is excellent with regard to a protective function of protecting electric wires from external damage. Accordingly, the charging cable can be appropriately regulated even in the event of a collision of the vehicle.

[0024] With the power unit room installing structure according to a fifth aspect of the disclosure, in any one aspect of the first to fourth aspects, [0025] the vehicle is a plug-in hybrid electric vehicle (PHEV) equipped with the power unit that includes an engine and a transaxle, [0026] in the power unit room, the engine and the transaxle are arrayed side by side in a vehicle width direction, [0027] the high voltage component is disposed between the engine and the dash panel, [0028] the charger is installed on the vehicle upward side from the transaxle, and also side by side with the engine in the vehicle width direction in plan view, [0029] the charging cable is routed so as to pass through a space between the engine and the charger and head toward the dash panel, and [0030] the regulating cable is configured to, by covering a portion of the charging cable from the vehicle upward side, regulate movement of the charging cable toward the vehicle upward side, and allow movement of the charging cable toward the dash panel side and also the high voltage component side.

[0031] In the power unit room installing structure according to the fifth aspect of the disclosure, the high voltage component is disposed between the engine and the dash panel, the transaxle is disposed side by side in the vehicle width direction with the engine, and the charger is disposed on the vehicle upward side of the transaxle. Thus, even in PHEVs having many components installed in the power unit room, routing the charging cable extending from the charger through the space between the engine and the charger enables the extra length of the charging cable to be efficiently secured. Further, causing the extra length of the charging cable to move to the dash panel side and also the high voltage component side by regulating by the regulating cable enables the high voltage component to be covered from the vehicle upward side. Thus, occurrence of a secondary accident such as electric shock or the like can be suppressed, while efficiently securing extra length of the charging cable.

[0032] As described above, according to the disclosure, occurrence of a secondary accident can be suppressed in preparation for damage of a high voltage component in the power unit room in the event of a collision.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0034] FIG. 1 is a schematic plan view schematically showing a power unit room installing structure according to the present embodiment;

[0035] FIG. 2 is an enlarged schematic plan view of the power unit room installing structure according to the present embodiment;

[0036] FIG. 3 is an enlarged schematic side view of the power unit room installing structure according to the present embodiment;

[0037] FIG. 4 is a schematic plan view showing an enlarged view of a power unit room installing structure according to the present embodiment in which barriers collide with each other from the front side of the vehicle; and

[0038] FIG. 5 is a schematic plan view illustrating an enlarged state in which an area collides with a power unit room installing structure as a comparative example from a vehicle front side.

DETAILED DESCRIPTION OF EMBODIMENTS

[0039] Hereinafter, an embodiment according to the disclosure will be described with reference to FIGS. 1 to 4. For convenience of explanation, an arrow UP shown in the drawings is defined as a vehicle upward direction, an arrow FR is defined as a vehicle forward direction, and an arrow LH is defined as a vehicle leftward direction. Thus, when terms indicating directions, i.e., upward and downward, forward and rearward, and right and left are used in the following description without any specification, these mean upward and downward of the vehicle, forward and rearward of the vehicle, and right and left of the vehicle. Further, the right-left direction is synonymous with a vehicle width direction.

Overall Configuration of the Power Unit Room

[0040] As illustrated in FIGS. 1 to 4, the power unit room installing structure **10** according to the present embodiment is mounted on the front portion of the vehicle **12**. A power unit room **R1** is provided at a front portion of the vehicle **12**, and a power unit **14** as a group of devices for generating power during traveling of the vehicle **12** is mounted. The power unit **14** is disposed to face the dash panel **18** that separates the power unit room **R1** and the vehicle cabin in the vehicle front-rear direction.

[0041] Note that, for example, the vehicle **12** is constituted by a chargeable plug-in hybrid electric vehicle (PHEV vehicle). Therefore, the power unit **14** includes an engine **20** as an internal combustion engine and a transaxle **22**. The transaxle **22** has a configuration in which, for example, a motor (not shown) as a drive source, an inverter (not shown) that controls electric power supplied to the motor, and a transmission (not shown) that appropriately decelerates the rotation of the motor and transmits it to the drive wheels are integrated.

[0042] In the power unit room **R1**, the engine **20** and the transaxle **22** are mounted side by side in a plan view. A charger **26** is mounted on the upper side of the transaxle **22**. The charger **26** is mounted on the upper side of the transaxle **22** and side by side in the vehicle width direction with respect to the engine **20** in a plan view. Furthermore, a charging cable **30** is arranged from the charger **26** towards the dash panel **18**. The tip of the charging cable **30** is routed to a charging port **28** for external power connection provided on the outer plate of the vehicle **12**.

[0043] For example, in the case of the vehicle **12** having the right steering wheel, the charging port **28** is provided at an upper portion of a left front fender (not shown) as an outer plate thereof. A battery **32** (see FIG. **1**) charged through the charger **26** is mounted at a substantially central portion of the vehicle **12** in the front-rear direction.

[0044] The charger **26** is formed in a substantially rectangular box shape having a predetermined thickness (low height) (see FIG. **3**). The charger **26** is fixed to the upper surface of the transaxle **22** via a support member (not shown) in such a manner that its longitudinal direction is in a posture along the front-rear direction. In addition, the transaxle **22** according to the present embodiment is disposed on the left side of the engine **20** in the power unit room **R1**.

[0045] Further, in the power unit room **R1**, a radiator **24** (see FIGS. **2** and **3**) and a cooling device (not shown) for cooling the charger **26** are disposed on the front side (the front end in the power unit room **R1**) of the engine **20** and the transaxle **22** (the charger **26**). The cooling device and the charger **26** are connected by a pipe (not shown) through which the refrigerant passes.

[0046] Here, the engine **20** and the transaxle **22** are supported by a pair of left and right front side members **40** via a mount (not shown). The front end portion of the front side member **40** is connected to the bumper reinforcement **42** extending in the vehicle width direction. The engine **20** and the transaxle **22** may be supported by a suspension member (not shown) attached to the lower surface of the front side member **40** via a mount.

[0047] Further, in the power unit room **R1**, a cowl **16** (see FIG. **3** and omitted in FIG. **2**) extending in the vehicle-width direction is disposed on the rear side of the engine **20** and the charger **26** (the rear end in the power unit room **R1**) in a plan view. The power unit room **R1** is covered from above by a substantially flat plate-shaped hood **13** (see FIG. **3**). That is, the hood **13** extending in the front-rear direction and the vehicle width direction is arranged above the engine **20** and the charger **26**.

High Voltage Component

[0048] As shown in FIG. **2**, a high voltage component **50** is disposed between the power unit **14** and the dash panel **18**. Here, for example, the high voltage component **50** is a water heater (so-called PTC heater) that functions as a heat source for warming the battery **32**. The high voltage component **50** heats the water instead of the engine waste heat, and circulates the hot water by a water pump (not shown) to warm the battery **32** mounted on the lower side of the floor of the vehicle cabin. Thus, in use in a cold district or the like, it is possible to prevent the battery **32** from being cooled down and shorten the charging time. The high voltage component **50** may be a water heater that functions as a heat source for heating the vehicle cabin.

[0049] Since the high voltage component **50** represented by the above-described water heater or the like is a high-voltage heater, it contributes to enhancing the safety of the occupant in the vehicle cabin by disposing the high-voltage heater between the power unit **14** and the dash panel **18**, that is, outside the vehicle cabin.

[0050] In the present embodiment, the high voltage component **50** is formed in a substantially rectangular box shape having a small thickness (low height) in the vehicle front-rear direction (see FIGS. **2** and **3**). The high voltage component **50** has a posture in which the longitudinal direction thereof is along the vehicle width direction, and is fixed to a substantially central portion of the dash panel **18** in the vehicle vertical direction via a support member (not shown). In addition, the high voltage component **50** according to the present embodiment is disposed on the rear side of the engine **20** in the power unit room **R1**. In the present embodiment, the high voltage component **50** is provided with a live part **52** in which a high-voltage harness **54** is connected to a side surface **50A** (a left side surface in FIG. **2**) disposed facing the transaxle **22**.

Charging Cable

[0051] Further, as shown in FIG. **2**, a space **S** (a gap that is a dead space) is formed between the upper portion of the engine **20** and the charger **26** (a portion opposed in the vehicle width direction) in a plan view, which is longer in the front-rear direction and wider in width (length along the

vehicle width direction) than the outer diameter of the charging cable **30**. The charging cable **30** in the present embodiment is routed through the space S.

[0052] Specifically, a connector **27** is provided at one end of the charging cable **30**, the other end of which is connected to the charging port **28**. The connector **27** is connected to, for example, a portion of the upper surface of the charger **26** that is substantially in the center in the front-rear direction and that is away from the engine **20**. The charging cable **30** is routed to the rear side toward the dash panel **18** from the side facing the engine **20** of the charger **26** in the vehicle width direction.

[0053] That is, the charging cable **30** is drawn out from the connector **27** to the engine **20** side along the vehicle width direction, and then bent rearward so as to pass through the space S, and is wired. Then, the charging cable **30** arranged through the space S to the rear side is bent downward on the front side of the cowl **16**, and the front is arranged along the dash panel **18**.

[0054] In this way, the charging cable **30** can be said to be in a state in which the portion drawn out from the connector **27** is arranged to the dash panel **18** side through the space S without passing through the upper surface of the charger **26**, so as to have a certain margin. Here, the deflection of the charging cable **30** (the extra-long portion **30A**) means that, for example, a substantially arc-shaped loop-shape is formed (see FIG. 3). Then, the intermediate portion of the charging cable **30** bent downward from the bent portion of the charging cable **30** (hereinafter sometimes referred to as “extra-long portion **30A**”) is fixed by a fixing metal fitting **19** (see FIG. 2) such as a clamp to the dash panel **18**.

Regulating Cable

[0055] Here, as shown in FIGS. 2 and 3, at a position overlapping at least a part of the power unit **14** in a plan view, a part of the charging cable **30** is covered from the upper side by the regulating cable **60**.

[0056] Specifically, the regulating cable **60** is, for example, a cable that connects an inverter (not shown) included in the transaxle **22** and the charger **26**. The regulating cable **60** is also referred to as a PN cable because each of the positive and negative electrodes of the battery **32** is connected to an inverter. The regulating cable **60** is composed of a high-voltage harness. The regulating cable **60** includes a shield layer for preventing electromagnetic noise from being affected by the surrounding electronic devices and signal lines, and a resin or metal exterior material provided for protecting the electric wires, and therefore has a relatively large cross-sectional area and is excellent in a protective function of protecting the electric wires from trauma.

[0057] A first PN connector (Positive Negative Connector) **62** is provided at one end of the regulating cable **60**, and the first PN connector **62** is connected to, for example, a front portion of the connector **27** on the upper surface of the charger **26**. Further, the other end of the regulating cable **60**, a second PN connector **64** is provided, the second PN connector **64**, for example, is connected to the vehicle-vertical substantially central portion of the rear surface of the transaxle **22** (the surface facing the dash panel **18**). The first PN connector **62** and the second PN connector **64** are two-pole high-voltage connectors corresponding to the positive electrode and the negative electrode.

[0058] The regulating cable **60** is routed to the rear side toward the dash panel **18** from the side facing the engine **20** of the charger **26** in the vehicle width direction. The middle portion arranged toward the dash panel **18** passes through the space S and covers the extra-long portion **30A** drawn out from the connector **27** of the charging cable **30** from above. The regulating cable **60** is connected to the rear surface of the transaxle by being bent downward by being pulled out from the space S ahead of the portion covering the charging cable **30**.

[0059] In the arranged state as described above, the regulating cable **60** has a relatively short line length, and is in a state of hardly bending. In other words, in the regulating cable **60**, the extra length of the cable is set to be shorter than the extra length of the charging cable **30**. Therefore, the regulating cable **60** regulates the upward movement of the charging cable **30**.

[0060] Further, in the charging cable **30**, the extra-long portion **30A** that does not overlap with the regulating cable **60** is bent in a substantially arc-shaped loop-shape protruding toward the engine-**20** side and the dash panel **18** side in a plan view. Therefore, the regulating cable **60** allows the charging cable **30** to move toward the engine **20** side and toward the dash panel **18** side. Therefore, when the charging cable **30** moves during a front collision of the vehicle, the high voltage component **50** can be covered from above the vehicle by the extra-long portion **30A** of the charging cable **30**.

Action and Effect

[0061] As described above, in the power unit room installing structure **10** according to the present embodiment, the high voltage component **50** is disposed between the power unit **14** and the dash panel **18**. Further, a charging cable **30** extending from the charger **26** disposed on the vehicle upper side of the power unit **14** is arranged toward the dash panel **18**. In addition, a regulating cable **60** that covers the charging cable **30** is disposed in the power unit room **R1**. The regulating cable **60** is configured to regulate movement of the charging cable **30** toward the vehicle upper side by covering a portion of the charging cable **30** from the vehicle upper side, and allow movement to at least the dash panel **18** side.

[0062] Here, as an example of the movement of the charging cable **30**, a case where the barrier **B** collides with the power unit room installing structure **10** according to the present embodiment from the vehicle front side will be described with reference to FIG. **4**. Note that FIG. **4** shows a state in which a collision load is input in a so-called offset collision in which a left portion (about $\frac{1}{2}$ of the vehicle width) collides with the barrier **B** in a state of being wrapped with respect to the center in the vehicle width direction.

[0063] As shown in FIG. **4**, when a collision load is inputted to the power unit room **R1**, the collision load is transmitted from the bumper reinforcement **42** to the front side member **40**. When the collision load is transmitted to the front side member **40**, the fragile portion (crushable zone) provided at the front end portion of the front side member **40** is compressively plastically deformed to absorb the collision load. Thereafter, by further input of the collision load, the front end portion of the front side member **40** is bent and deformed inward in the vehicle width direction starting from the fragile portion, and abuts against the front end portion of the power unit **14** (transaxle **22**) and the end portion outside in the vehicle width direction. Accordingly, a collision load obliquely rearward toward the vehicle rear side and the anti-collision side (the vehicle width direction inner side) is input to the transaxle **22**. As a result, the transaxle **22** and the charger **26** disposed on the upper side of the transaxle **22** move obliquely rearward toward the vehicle rear side and the anti-collision side (the vehicle width direction inner side).

[0064] Here, as a comparative example, for example, as shown in FIG. **5**, a part of the charging cable **30** may not be covered with the regulating cable **60**. In this case, the extra-long portion **30A** of the charging cable **30** moves toward the vehicle upper side, and has a substantially arcuate loop-shaped behavior that protrudes toward the vehicle upper side and bends. Therefore, the vehicle upper side of the high voltage component **50** becomes an open space, and contact with the high voltage component **50** is facilitated.

[0065] On the other hand, in the present embodiment, the extra-long portion **30A** of the charging cable **30** is allowed to move toward the engine-**20** side and the dash panel **18** side by the restriction of the regulating cable **60**. Therefore, it moves between the power unit and the dash panel **18** to cover the high voltage component **50** from the vehicle upper side. Thus, even when the high voltage component **50** is damaged due to damage at the time of collision, the charging cable **30** is covered from the upper side. As a result, a person who opens the power unit room **R1** cannot easily touch the high voltage component **50**, and thus the generation of a secondary accident such as an electric shock is suppressed.

[0066] Further, as shown in FIG. **4**, in the present embodiment, the extra-long portion **30A** of the charging cable **30** is allowed to move so as to overlap with the live part **52** of the high voltage

component **50**. Therefore, it is possible to effectively suppress a person who opens the power unit room **R1** from contacting the live part **52** having a higher risk of electric shock.

[0067] In FIG. **4**, the offset collision of the left portion of the vehicle **12** has been described as an example. Similar effects can be obtained by moving the charging cable **30** toward the dash panel **18** in the case of an offset collision of a portion on the right side of the vehicle **12** or a so-called full-lap collision.

[0068] Further, in the present embodiment, the regulating cable **60**, the extra length of the cable, since the cable is set to be shorter than the extra length of the charging cable **30**, it is possible to easily restrict the movement of the charging cable to the vehicle upper side. Further, as compared with the case of restricting the moving of the charging cable using a fixed bracket such as clamping, since it does not require additional components, with reducing the number of components in the power unit room **R1**, it is possible to suppress the weight increment.

[0069] Further, in the present embodiment, a cable connecting the transaxle **22** and the charger **26** is used as the regulating cable **60**. The cable connecting the transaxle **22** and the charger **26** is usually composed of a high-voltage harness, and therefore has a relatively large cross-sectional area and is excellent in a protective function of protecting the electric wire from trauma. Therefore, it is possible to appropriately regulate the charging cable **30** even when the vehicle **12** collides.

[0070] Further, in the present embodiment, the vehicle **12** is a PHEV vehicle, and the high voltage component **50** is disposed between the engine **20** and the dash panel **18**. Further, the transaxle **22** is arranged side by side in the vehicle width direction with the engine **20**, and the charger **26** is arranged on the vehicle upper side of the transaxle **22**. In this way, even in PHEV vehicles having many components mounted in the power unit room, the charging cable **30** extending from the charger **26** is arranged through the space **S** between the engine **20** and the charger **26**. This makes it possible to efficiently secure **30A** of the extra-long portion of the charging cable **30**. Then, by regulating the regulating cable **60** to move the extra length of the charging cable **30** to the dash panel **18** side and the high voltage component **50** side, it is possible to cover the high voltage component from the vehicle upper side. In this way, in the power unit room installing structure **10** according to the present embodiment, it is possible to efficiently secure **30A** of the extra length portion of the charging cable **30**, and to suppress the generation of a secondary accident such as an electric shock.

[0071] The power unit room installing structure **10** according to the present embodiment has been described above. The power unit room installing structure according to the disclosure is not limited to the illustrated one, and can be appropriately changed in design without departing from the gist of the disclosure.

[0072] For example, in the above-described embodiment, the power unit room installing structure **10** is configured to be applied to the front portion of the vehicle, but may be applied to the rear portion of the vehicle.

[0073] Further, in the above embodiment, the power unit room installing structure **10** is applied to PHEV vehicles, but the power unit room installing structure **10** may be applied to a battery electric vehicle (EV vehicle). In this case, the power unit may be constituted by a transaxle in which a motor as a drive source, an inverter that controls electric power supplied to the motor, and a transmission that appropriately decelerates the rotation of the motor and transmits the rotation to the drive wheels are integrated.

Claims

1. A power unit room installing structure, comprising: a power unit that is installed in a power unit room of a vehicle, and that is disposed facing a dash panel separating the power unit room and a vehicle cabin in a vehicle front-rear direction; a high voltage component that is disposed between the power unit and the dash panel; a charger that is disposed on a vehicle upward side from the

power unit; a charging cable that is routed from the charger toward the dash panel; and a regulating cable that is another cable different from the charging cable, and that is configured to, by covering a portion of the charging cable from the vehicle upward side, regulate movement of the charging cable toward the vehicle upward side, and allow movement of the charging cable toward at least the dash panel side.

2. The power unit room installing structure according to claim 1, wherein the regulating cable is configured to allow movement of the charging cable toward the dash panel side such that the charging cable overlaps a live part of the high voltage component in plan view.

3. The power unit room installing structure according to claim 1, wherein the regulating cable is a cable in which an extra length of the cable is set to be shorter than an extra length of the charging cable.

4. The power unit room installing structure according to claim 1, wherein the regulating cable is a cable that connects a transaxle included in the power unit, and the charger.

5. The power unit room installing structure according to claim 1, wherein: the vehicle is a plug-in hybrid electric vehicle (PHEV) equipped with the power unit that includes an engine and a transaxle; in the power unit room, the engine and the transaxle are arrayed side by side in a vehicle width direction; the high voltage component is disposed between the engine and the dash panel; the charger is installed on the vehicle upward side from the transaxle, and also side by side with the engine in the vehicle width direction in plan view; the charging cable is routed so as to pass through a space between the engine and the charger and head toward the dash panel; and the regulating cable is configured to, by covering a portion of the charging cable from the vehicle upward side, regulate movement of the charging cable toward the vehicle upward side, and allow movement of the charging cable toward the dash panel side and also the high voltage component side.
